Imbalance correction reverses CNN-transformer rankings in melanoma classification: a controlled benchmark and fairness analysis

Nursultan Zhanat
Supervisor: Prof. Sun Baoshan

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Abstract

Deep learning models for melanoma detection are trained on imbalanced, predominantly light-skinned data. We ask whether modern imbalance correction changes the ranking of CNN and transformer architectures, and whether the resulting models perform fairly across skin tone strata. We benchmark ResNet-50, EfficientNet-B0, and ViT-S/16 on HAM10000 under three pre-declared conditions: cross-entropy with standard augmentation, focal loss with class-balanced sampling, and cross-entropy with strong augmentation. Each condition uses three seeds, a lesion-aware split, and a validation-selected threshold at 95% specificity. Focal loss with class-balanced sampling improves ViT-S/16 by 8.2 sensitivity points but reduces ResNet-50 by 11.7 and EfficientNet-B0 by 7.9, which reverses the architecture ranking. The AUC ranking is stable. A lightness-stratified analysis shows that the darkest stratum has the lowest AUC and the highest false positive rate in all nine configurations, but only 43 test images and six melanomas fall in that stratum, so the sensitivity estimate is underpowered. In an EfficientNet-B0 pilot, attribution mass inside the lesion decreased with darker lightness group ($\rho = -0.17$, $p < 10^{-7}$), which is consistent with more attribution leakage on darker images. The results show that imbalance correction is not neutral across architectures and that dark-skin dermoscopic data remain the main bottleneck for a reliable fairness claim.

1 Introduction

Melanoma is the most lethal form of skin cancer, and survival depends on early detection. Deep learning models have reached dermatologist-level accuracy on public dermoscopic benchmarks such as HAM10000 and the ISIC challenges [14, 4]. Three problems remain.

Public datasets are imbalanced. Benign lesions outnumber melanomas by roughly ten to one, and a classifier that predicts the majority class can look accurate while missing cancers. Focal loss and class-balanced sampling are common remedies [9], but their effect is often reported for one architecture and one dataset.

The datasets are also skewed toward lighter skin. Fitzpatrick skin types V and VI are rare in public dermoscopic data, and models trained on lighter skin can lose sensitivity on darker skin [7, 5, 11]. The data needed to measure the gap are themselves scarce.

Deep models are opaque as well. Gradient-based attribution methods such as Grad-CAM and Integrated Gradients can show which pixels drive a prediction [12, 13], but most dermatology papers use them as illustrations. They are rarely turned into quantitative measurements that can be tested across skin tone groups.

Most studies examine one of these problems at a time. Architecture comparisons often apply a single imbalance correction without checking whether it is neutral across architectures, and fairness studies report group-level performance without connecting it to attribution behaviour. This study connects the three in one benchmark.

We study HAM10000 [14] with a binary melanoma endpoint and three architectures: ResNet-50, EfficientNet-B0, and ViT-S/16. We use a lesion-aware split, three seeds per condition, and a validation-selected threshold at 95% specificity. The test set is touched once per run.

Our contributions are:

1. A controlled benchmark of three architectures under three pre-declared conditions: CE baseline, focal loss with class-balanced sampling, and CE with strong augmentation.

2. Evidence that focal loss with class-balanced sampling improves ViT-S/16 but reduces the sensitivity of both CNNs, which reverses the architecture ranking on sensitivity at 95% specificity. The AUC ranking is stable.

3. A lightness-stratified fairness analysis that quantifies the representation gap and a consistent dark-stratum AUC and false positive rate signal, while showing that the dark-stratum sensitivity estimate is underpowered.

4. A quantitative XAI protocol for attribution leakage and faithfulness, used to test whether attribution mass outside the lesion predicts errors and tone group. The XAI results are reported in Section 4.5.

This is a controlled pilot on one public dataset. The lightness strata are image-based measurements, not clinical Fitzpatrick labels, and the results do not establish clinical utility. The external tone-labeled validation is reported in Section 4.6.

2 Related work

2.1 Melanoma classification and public datasets

Public dermoscopic datasets have driven most of the recent progress. HAM10000 contains 10,015 images across seven diagnostic classes [14], and the ISIC challenges have expanded the scale and the number of sources [4]. Models based on ResNet, EfficientNet, ConvNeXt, ViT, and Swin have all been reported at or above dermatologist-level accuracy on these benchmarks. The comparisons usually differ in preprocessing, split protocol, and threshold choice, which makes cross-paper rankings difficult to interpret.

2.2 Class imbalance

Focal loss down-weights easy examples and is the most common loss-level remedy for class imbalance [9]. Class-balanced sampling and class-weighted cross-entropy are common data-level and loss-level alternatives. Strong photometric and geometric augmentation is often added at the same time. The three interventions are usually evaluated together, so their separate effects and their interactions with architecture are rarely reported.

2.3 Fairness in dermatology

Fitzpatrick17k provided the first large clinical image set with Fitzpatrick labels and showed performance gaps across skin types [7]. The Diverse Dermatology Images dataset added biopsy-proven clinical images with balanced skin tone representation [5]. SCIN added crowdsourced clinical images with self-reported and estimated Fitzpatrick labels [6]. Pipsqueak collected dermoscopic melanocytic lesions in darker skin and reported a large sensitivity drop on Fitzpatrick V-VI [11]. A 2025 systematic review identifies fairness, generalizability, and replicability as the main unresolved problems in melanoma AI [1].

The dermoscopic datasets with Fitzpatrick labels remain small. HIBA contains 1,616 images from Argentina with Fitzpatrick labels [8], and PROVe-AI contains 603 biopsy-confirmed lesions [10]. Neither includes Fitzpatrick V-VI. This limits what can be claimed from dermoscopic data alone.

2.4 Explainable AI in dermatology

Grad-CAM and Integrated Gradients are the most used attribution methods in dermatology [12, 13]. Clinician-facing studies show that explanations can improve trust and diagnostic accuracy [3], but the explanations are usually evaluated qualitatively. Quantitative localization and faithfulness metrics are more common in the general XAI literature than in dermatology.

2.5 Quantitative XAI metrics

Attribution localization can be measured against lesion segmentation masks: the fraction of attribution mass inside the lesion, the pointing game, and the attribution IoU. Faithfulness can be measured with deletion and insertion curves, infidelity, and sensitivity-n. These metrics make it possible to test whether an explanation is clinically meaningful rather than visually plausible.

2.6 Gap

Three questions remain open: whether imbalance correction changes the ranking of architecture families, whether the correction affects skin tone groups differently, and whether attribution metrics can detect a group-level problem before it appears in aggregate sensitivity. We address the first two and define the protocol for the third.

3 Methods

3.1 Data and splits

We use HAM10000, which contains 10,015 dermoscopic images from 7,470 lesions across seven diagnostic classes [14]. The task is binary: melanoma versus all other diagnoses. The dataset contains 1,113 melanoma images, 11.1% of the total.

We use a lesion-aware split so that images of the same lesion never appear in more than one split. The split is 70% training, 15% validation, and 15% test. The test set contains 1,527 images and 186 melanomas. The split is generated once and reused by every run.

3.2 Preprocessing and augmentation

Images are resized to 160 pixels for the Stage 1 screen and 224 pixels for the Stage 2 comparison, and normalized with ImageNet statistics. The standard augmentation pipeline applies horizontal

and vertical flips, random 90-degree rotations, and mild color jitter. The strong pipeline increases the color jitter and adds affine transformations with scale, translation, and rotation. The strong regime is designed to simulate domain shift.

3.3 Architectures

We compare ResNet-50, EfficientNet-B0, and ViT-S/16. All models use ImageNet pretrained weights and a two-class output. The ViT uses dynamic image sizing so that it accepts the 224-pixel input. The three models cover a standard CNN, an efficient CNN, and a transformer.

3.4 Training

All models use AdamW with a learning rate of 3×10^{-4}, weight decay 1×10^{-4}, and a cosine schedule. The batch size is 64, the maximum number of epochs is 15, and early stopping uses a patience of 5 on validation balanced accuracy. Gradients are clipped to a norm of 1.0. Training uses mixed precision when a GPU is available.

3.5 Conditions

Stage 1 is a screen on EfficientNet-B0 at 160 pixels. It contains eight configurations: CE with a random sampler, CE with class-balanced sampling, weighted CE with a random sampler, focal loss with $\gamma = 2$ and a random sampler, focal loss with $\gamma = 2$ and class-balanced sampling, focal loss with $\gamma = 3$ and class-balanced sampling, and two strong-augmentation variants. Each configuration uses three seeds, for 24 runs.

Stage 2 compares three pre-declared conditions across the three architectures at 224 pixels:

1. CE baseline: cross-entropy loss, random sampler, standard augmentation.
2. Planned imbalance correction: focal loss with $\gamma = 2$, class-balanced sampling, standard augmentation.
3. Stage 1 winner: cross-entropy loss, random sampler, strong augmentation.

Each condition uses three seeds, for 27 runs. The Stage 2 conditions are fixed before the test results are seen.

3.6 Metrics and threshold

We report sensitivity, specificity, balanced accuracy, AUC, and sensitivity at 95% specificity. The operating threshold is selected on the validation split at 95% specificity, frozen, and applied once to the test split. Each run writes per-sample test predictions so that the same scores can be reused for fairness and XAI analysis.

3.7 Statistical analysis

We report the mean and standard deviation across three seeds. We add 1,000-sample bootstrap confidence intervals on test for sensitivity, specificity, and AUC. The bootstrap resamples images within a run and skips degenerate samples. The intervals describe test-set uncertainty. The seed standard deviation describes run-to-run variation.

3.8 Fairness analysis

HAM10000 has no Fitzpatrick labels. We therefore compute an image-based lightness proxy from non-lesional skin. The image is converted to CIELAB, the lesion mask is dilated by 11 pixels and excluded, and very dark pixels ($L^* < 10$) and very bright pixels ($L^* > 95$) are removed. The median L^* of the remaining skin pixels is the proxy.

We group images into light ($L^* > 70$), medium ($55 < L^* \leq 70$), and dark ($L^* \leq 55$). These thresholds are specific to the HAM10000 lightness distribution. They are image-based strata, not clinical Fitzpatrick labels.

We also computed the Individual Typology Angle (ITA), $\text{ITA} = \arctan((L^* - 50)/b^*) \times 180/\pi$. The ITA was unstable on 37% of images because the b^* channel was near zero or negative after dermatoscope color casts. The L^* proxy is therefore the primary analysis, and the ITA is treated as a sensitivity check.

For each Stage 2 run, we compute per-group sensitivity, specificity, false positive rate, false negative rate, AUC, and bootstrap intervals. We report the maximum group gap and the light-to-dark sensitivity ratio.

External validation uses HIBA [8] and PROVe-AI [10], which are dermoscopic datasets with Fitzpatrick labels. Both contain Fitzpatrick I-IV only. A clinical dark-skin dataset, such as DDI [5] or PASSION [2], is used for the Fitzpatrick V-VI comparison, with the imaging modality difference stated as a limitation.

3.9 XAI analysis

For each selected run we compute Grad-CAM heatmaps for the test images. We measure:

- Attribution mass inside the lesion: $M = \sum_{i \in \text{lesion}} h_i / \sum_i h_i$, where h_i is the normalized heatmap value at pixel i.
- Pointing game accuracy: the fraction of images where the maximum heatmap value falls inside the lesion mask.
- Attribution IoU: the intersection over union between the thresholded heatmap and the lesion mask.
- Deletion and insertion AUC: the area under the target-class probability curve when pixels are removed or added in descending heatmap order.

We then compute the Spearman correlation between attribution mass and error status, and between attribution mass and lightness group. The XAI analysis uses the same test split and the same per-sample predictions as the classification and fairness analyses.

3.10 Reproducibility

Every run writes its configuration, git commit, seed, device, package versions, metrics, and per-sample predictions. The split is cached and reused. The code, configs, and table generation scripts are available in the project repository.

Table 1: Stage 1 screen on EfficientNet-B0 at 160 pixels. Validation sensitivity is used for model selection. Test sensitivity is reported at the validation-selected 95% specificity threshold. Values are mean and standard deviation across three seeds.

Configuration	Val sens	Test sens (95% CI)	Specificity	Balanced acc	AUC
stage1_ce_random_strong	0.556 ± 0.027	0.513 (0.438 to 0.582)	0.943	0.728	0.901
stage1_ce_random	0.536 ± 0.016	0.530 (0.456 to 0.600)	0.948	0.739	0.897
stage1_ce_cb	0.521 ± 0.086	0.484 (0.411 to 0.554)	0.938	0.711	0.890
stage1_focal2_random	0.509 ± 0.041	0.487 (0.412 to 0.560)	0.946	0.717	0.894
stage1_focal2_cb	0.502 ± 0.039	0.480 (0.408 to 0.551)	0.943	0.712	0.897
stage1_wce_random	0.502 ± 0.043	0.493 (0.421 to 0.564)	0.937	0.715	0.896
stage1_focal2_cb_strong	0.483 ± 0.013	0.491 (0.421 to 0.558)	0.943	0.717	0.896
stage1_focal3_cb	0.449 ± 0.051	0.478 (0.405 to 0.553)	0.946	0.712	0.899

4 Results

4.1 Stage 1: loss, sampling, and augmentation screen

Table 1 reports the Stage 1 screen on EfficientNet-B0 at 160 pixels. The validation winner is CE with strong augmentation, with a validation sensitivity of 0.556 ± 0.027 at 95% specificity. The CE baseline reaches 0.536 ± 0.016. The planned focal loss with class-balanced sampling reaches 0.502 ± 0.039, and focal loss with $\gamma = 3$ and class-balanced sampling reaches 0.449 ± 0.051.

The planned imbalance correction did not beat the CE baseline on validation. The difference between the CE baseline and the strong-augmentation variant is small relative to the seed standard deviation. The test columns in Table 1 are descriptive; the model selection used validation only.

4.2 Stage 2: architecture comparison

Table 2 reports the Stage 2 comparison at 224 pixels. The three conditions are the CE baseline, focal loss with class-balanced sampling, and CE with strong augmentation.

At the CE baseline, ResNet-50 reaches a test sensitivity of 0.570 (95% CI 0.497 to 0.639), EfficientNet-B0 reaches 0.556 (0.480 to 0.624), and ViT-S/16 reaches 0.400 (0.329 to 0.471). The two CNNs lead the transformer by a wide margin.

Under focal loss with class-balanced sampling, ResNet-50 falls to 0.453 (0.379 to 0.523), EfficientNet-B0 falls to 0.477 (0.405 to 0.548), and ViT-S/16 rises to 0.482 (0.408 to 0.556). The correction helps the transformer by 8.2 points and hurts the CNNs by 11.7 and 7.9 points.

The architecture ranking therefore reverses. At the CE baseline the order is ResNet-50, EfficientNet-B0, ViT-S/16. Under focal loss with class-balanced sampling the order is ViT-S/16, EfficientNet-B0, ResNet-50.

Strong augmentation is close to neutral. ResNet-50 reaches 0.568 (0.498 to 0.637), EfficientNet-B0 reaches 0.563 (0.489 to 0.636), and ViT-S/16 reaches 0.435 (0.366 to 0.503). The intervals overlap with the CE baseline for the two CNNs.

4.3 AUC is stable while sensitivity ranking flips

The AUC ranking does not follow the sensitivity ranking. At the CE baseline, the AUC is 0.914 for EfficientNet-B0, 0.897 for ResNet-50, and 0.854 for ViT-S/16. Under focal loss with class-balanced sampling, the AUC is 0.891 for EfficientNet-B0, 0.884 for ResNet-50, and 0.872 for ViT-S/16. EfficientNet-B0 remains first and ViT-S/16 remains last in both conditions.

Table 2: Stage 2 comparison at 224 pixels. Test sensitivity is reported at the validation-selected 95% specificity threshold. Values are mean and standard deviation across three seeds.

Configuration	Val sens	Test sens (95% CI)	Specificity	Balanced acc	AUC
stage2_efficientnet_b0_ce_random_strong	0.521 ± 0.021	0.563 (0.489 to 0.636)	0.952	0.757	0.912
stage2_efficientnet_b0_ce_random	0.553 ± 0.035	0.556 (0.480 to 0.624)	0.953	0.754	0.914
stage2_efficientnet_b0_focal2_cb	0.481 ± 0.042	0.477 (0.405 to 0.548)	0.950	0.713	0.891
stage2_resnet50_ce_random	0.526 ± 0.044	0.570 (0.497 to 0.639)	0.951	0.761	0.897
stage2_resnet50_ce_random_strong	0.549 ± 0.015	0.568 (0.498 to 0.637)	0.942	0.755	0.899
stage2_resnet50_focal2_cb	0.489 ± 0.027	0.453 (0.379 to 0.523)	0.947	0.700	0.884
stage2_vit_s_focal2_cb	0.474 ± 0.034	0.482 (0.408 to 0.556)	0.940	0.711	0.872
stage2_vit_s_ce_random_strong	0.466 ± 0.030	0.435 (0.366 to 0.503)	0.941	0.688	0.866
stage2_vit_s_ce_random	0.423 ± 0.042	0.400 (0.329 to 0.471)	0.941	0.670	0.854

Table 3: Lightness-stratified test sensitivity at 95% specificity. Values are mean and standard deviation across three seeds. The dark stratum contains 43 test images and six melanoma cases. These are image-based lightness groups, not clinical Fitzpatrick labels.

Configuration	Light sens	Medium sens	Dark sens	Dark AUC	Dark FPR	Dark n / pos.
EfficientNet-B0 / CE	0.477 ± 0.031	0.609 ± 0.050	0.722 ± 0.096	0.871	0.117	43 / 6
EfficientNet-B0 / CE+strong	0.519 ± 0.054	0.599 ± 0.086	0.556 ± 0.096	0.833	0.090	43 / 6
EfficientNet-B0 / Focal+CB	0.420 ± 0.064	0.522 ± 0.076	0.500 ± 0.167	0.806	0.081	43 / 6
ResNet-50 / CE	0.502 ± 0.036	0.633 ± 0.050	0.444 ± 0.096	0.803	0.117	43 / 6
ResNet-50 / CE+strong	0.527 ± 0.026	0.603 ± 0.038	0.556 ± 0.096	0.854	0.135	43 / 6
ResNet-50 / Focal+CB	0.399 ± 0.093	0.481 ± 0.066	0.722 ± 0.192	0.848	0.162	43 / 6
ViT-S/16 / CE	0.342 ± 0.091	0.451 ± 0.065	0.333 ± 0.167	0.740	0.144	43 / 6
ViT-S/16 / CE+strong	0.407 ± 0.054	0.471 ± 0.132	0.222 ± 0.096	0.763	0.126	43 / 6
ViT-S/16 / Focal+CB	0.362 ± 0.062	0.586 ± 0.010	0.389 ± 0.096	0.712	0.189	43 / 6

The stability of the AUC ranking suggests that the sensitivity reversal is specific to the thresholded operating point. Calibration and threshold placement are plausible mechanisms. Both metrics matter: a clinical screening claim depends on the thresholded sensitivity, and the AUC describes the ranking quality of the scores.

4.4 Fairness across lightness strata

The test set contains 644 light images with 81 melanomas, 840 medium images with 99 melanomas, and 43 dark images with 6 melanomas. The dark stratum is 2.8% of the test set and 3.2% of the melanoma cases.

Table 3 reports the per-group sensitivity and the dark-stratum AUC and false positive rate for the nine Stage 2 configurations.

The dark stratum has the lowest AUC in all nine configurations. The dark AUC ranges from 0.71 to 0.87, against 0.83 to 0.90 for the light stratum and 0.88 to 0.93 for the medium stratum.

It also has the highest false positive rate. The dark false positive rate ranges from 0.08 to 0.19, against 0.04 to 0.07 for the light and medium strata.

The dark-stratum sensitivity is too uncertain to support a strong claim. With six melanoma cases, the estimates range from 0.22 to 0.72 and the bootstrap intervals span 0 to 1.

The medium stratum has the highest sensitivity in most configurations. The light stratum is often lower, which is consistent with a contrast effect: pale lesions on pale skin are harder to separate. Lesion contrast, not only skin tone, appears to drive difficulty.

Table 4: EfficientNet-B0 XAI pilot on the HAM10000 test split. Values are means across three seeds. The dark stratum contains 129 images and 18 melanoma cases in this sample.

Tone group	n	n positive	Attribution mass	Pointing game	Attribution IoU	Error rate
light	450	36	0.456	0.718	0.333	0.076
medium	450	54	0.374	0.613	0.284	0.091
dark	129	18	0.342	0.449	0.208	0.140

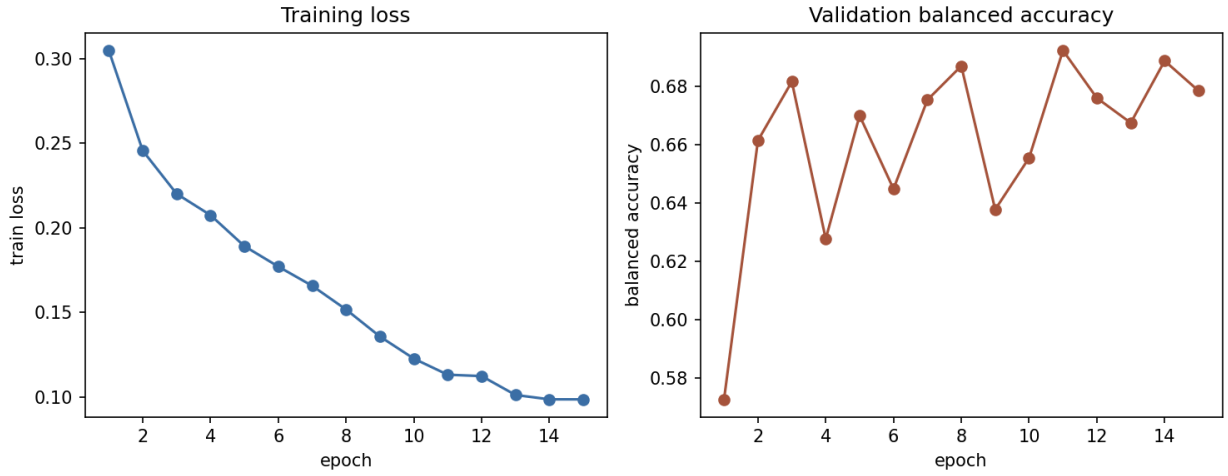


Figure 1: Training loss and validation balanced accuracy for the median seed of ResNet-50 with the CE baseline. The model trains for 15 epochs and restores the best validation checkpoint.

4.5 Quantitative XAI

The analysis measures attribution mass inside the lesion, pointing game accuracy, attribution IoU, and deletion and insertion AUC for the selected Stage 2 runs. The protocol is defined in Section 3.9.

In the EfficientNet-B0 pilot (three seeds, 1,027 test images), the dark stratum had lower attribution mass inside the lesion, lower pointing-game accuracy, and lower attribution IoU than the light and medium strata (Table 4). The Spearman correlation between attribution mass and tone group was -0.170 ($p < 10^{-7}$), so darker images received less attribution inside the lesion. The correlation between attribution mass and error was small and not significant ($\rho = -0.023$, $p = 0.46$). The ResNet-50 and ViT-S/16 analyses are in progress.

4.6 External tone-labeled validation

The external validation compares HIBA and PROVe-AI for Fitzpatrick I-IV and a clinical dark-skin dataset for Fitzpatrick V-VI. HIBA and PROVe-AI contain no Fitzpatrick V-VI images. The clinical comparison uses DDI or PASSION, with the imaging modality difference reported as a limitation.

4.7 Figures

Figure 1 shows the training curves for the median seed of the best Stage 2 configuration by mean test sensitivity, ResNet-50 with the CE baseline. Figure 2 shows the corresponding test ROC curve.

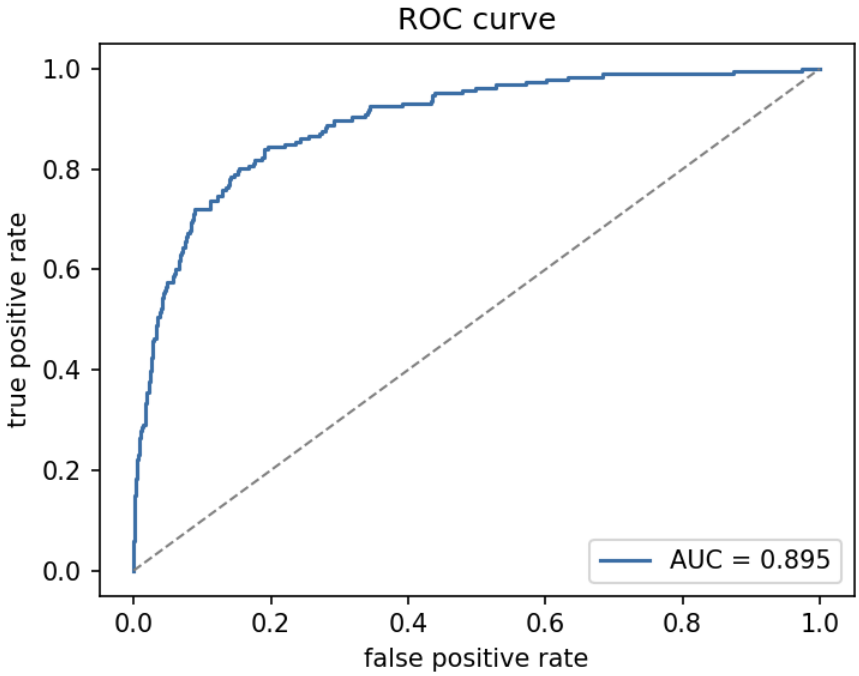


Figure 2: Test ROC curve for the median seed of ResNet-50 with the CE baseline. The operating threshold is selected on validation at 95% specificity.

5 Discussion

5.1 Imbalance correction is architecture-dependent

Focal loss with class-balanced sampling is not a neutral intervention. It helps the transformer and hurts both CNNs. A benchmark that applied the correction to one architecture and generalized the result would have reached the wrong conclusion about the others.

The effect has a plausible mechanism. Focal loss changes the gradient distribution, and class-balanced sampling changes the effective class prior. A transformer with weaker inductive bias may benefit from the harder examples and the reweighted prior, while a CNN with strong local features may overfit the minority samples or lose calibration. The experiment cannot separate those mechanisms. We report the effect as an empirical interaction, not a causal explanation.

5.2 The ranking flip and the stable AUC

The sensitivity ranking reverses, but the AUC ranking does not. EfficientNet-B0 remains first and ViT-S/16 remains last in both the baseline and corrected conditions. The reversal is therefore specific to the thresholded operating point.

This matters for how model comparisons are reported. A single threshold at 0.5, or a single sensitivity number, can make an architecture look better or worse depending on the correction. Reporting sensitivity at a fixed specificity, AUC, and calibration together gives a more complete picture. We leave a full calibration analysis to future work.

5.3 The fairness signal

The dark stratum has the lowest AUC and the highest false positive rate in all nine configurations. The direction is consistent even though the sample is small. The most likely interpretation is that the model's ranking quality degrades on darker images, which produces more false positives at the fixed specificity threshold.

The dark-stratum sensitivity is underpowered. Six melanoma cases cannot support a precise estimate, and the bootstrap intervals span the full range. We therefore treat the AUC and false positive rate as the more stable signals and the sensitivity estimate as inconclusive.

The light stratum is often harder than the medium stratum. This is consistent with a contrast effect: a pale lesion on pale skin has a lower boundary contrast, and the model may attend to the surrounding skin. Skin tone is not the only source of difficulty. Lesion contrast, hair, rulers, and illumination can all change the model's attention.

5.4 The data bottleneck

The external dermoscopic datasets with Fitzpatrick labels, HIBA and PROVe-AI, contain Fitzpatrick I-IV only. Pipsqueak, the only dermoscopic dark-skin set, has 16 core images. The clinical datasets with Fitzpatrick V-VI, such as DDI, SCIN, and PASSION, differ from HAM10000 in imaging modality.

This is the main obstacle to a strong dark-skin claim. This study can quantify the representation gap, report the consistent dark-stratum AUC and false-positive signal, and show the limitation of the available dermoscopic data. A precise dark-skin sensitivity estimate requires new dermoscopic data collection or a carefully controlled cross-modality study.

5.5 What XAI adds

The classification and fairness results describe what the model does. The XAI analysis tests whether it attends to the lesion or to the surrounding skin. If attribution mass outside the lesion is higher in the dark stratum and correlates with errors, attribution leakage becomes an early warning signal that can be measured before group-level sensitivity collapses. The protocol is defined in Section 3.9.

5.6 Implications for model selection

The choice of architecture depends on the correction. At the CE baseline, ResNet-50 has the highest sensitivity at 95% specificity. Under focal loss with class-balanced sampling, ViT-S/16 is the best. A deployment decision that ignores the training condition can pick the wrong model.

For the fairness analysis, this means the evaluation should include the architecture-condition combinations, not a single global best model. The best model on aggregate sensitivity may not be the best model in the dark stratum, and the ranking may change again after the external validation.

6 Limitations

- We use HAM10000 only. Cross-dataset generalization to PH2, BCN20000, or ISIC 2020 is not yet measured.
- The endpoint is binary melanoma versus all other diagnoses. Seven-class classification is a secondary experiment.

- Each condition uses three seeds. The seed standard deviation is reported, but more seeds would tighten the estimates.
- Stage 1 uses 160-pixel images and Stage 2 uses 224-pixel images. The two stages are not directly comparable.
- The fairness analysis uses an image-based lightness proxy, not clinical Fitzpatrick labels. The proxy is sensitive to illumination, color calibration, and lesion contrast.
- The dark stratum contains 43 test images and six melanomas. The sensitivity estimate is underpowered, and the dark-stratum results are a signal rather than a precise effect size.
- HIBA and PROVe-AI contain Fitzpatrick I-IV only. The clinical datasets with Fitzpatrick V-VI differ in imaging modality, so the dark-skin comparison is confounded.
- No clinician reader study was conducted, so the clinical relevance of the attribution maps is not yet validated with dermatologists.
- No deployed system, prospective validation, or regulatory analysis is included.

7 Ethics and reproducibility

All data are public, de-identified research datasets. HAM10000 and the ISIC archive are released for research use, and the external datasets are used under their respective licenses. The study involves no patient contact and no new data collection.

The system is a research prototype, not a diagnostic device. The results do not establish clinical utility.

The Fitzpatrick scale has known limitations: it is a coarse phototype scale, it does not capture the full range of skin tones, and it conflates pigmentation with UV response. We use it because the datasets and the fairness literature use it. The lightness strata are image-based measurements, not clinical labels.

The representation gap is a finding with an ethical consequence: the data needed to measure dark-skin performance are scarce. We report the small number of dark-skin dermoscopic images. Synthetic dark-skin images are not used to claim fairness improvements.

The code, configs, splits, per-sample predictions, and table generation scripts are released with the paper. The repository records the git commit, seed, device, and package versions for every run.

8 Conclusion

We benchmarked ResNet-50, EfficientNet-B0, and ViT-S/16 on HAM10000 under three pre-declared conditions. Focal loss with class-balanced sampling improved the transformer but reduced the sensitivity of both CNNs, which reversed the architecture ranking on sensitivity at 95% specificity. The AUC ranking stayed stable, which points to calibration and threshold effects rather than a change in ranking quality.

The fairness analysis showed that the darkest lightness stratum had the lowest AUC and the highest false positive rate in all nine configurations. The stratum contains only 43 test images and six melanomas, so the sensitivity estimate is underpowered. The external dermoscopic datasets

with Fitzpatrick labels contain no Fitzpatrick V-VI images, which is the main obstacle to a precise dark-skin claim.

The next steps are the quantitative XAI analysis, the HIBA and PROVe-AI external validation, and a clinical dark-skin evaluation with DDI or PASSION. The combined results will test whether attribution leakage can detect the group-level problem before it appears in aggregate sensitivity.

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